FinSecure Pay - End-to-End Architecture

Git -> Jenkins CI/CD with security gates -> Docker Hub -> Kubernetes (AWS EKS) with monitoring, logging and alerting

CI/CD pipeline (Jenkins, pipeline-as-code: Jenkinsfile)

Developer
git push

GitHub
badri35/capstone

Jenkins
polls Git every minute

Build + Test
Maven, JUnit 5 (19 tests)

Security checks
SpotBugs + Trivy deps/secrets

Quality gate
HIGH/CRITICAL or failed test = STOP

Docker build
multi-arch amd64 + arm64, non-root

Image scan
Trivy image: HIGH/CRITICAL blocks

Docker Hub
badri3590/finsecure-pay

Deploy
kubectl set image

Verify
rollout status

Rollback
rollout undo

if rollout fails

Secrets: Jenkins credential store (masked) - none in git | Email on every build via SMTP

deploy new version

AWS EKS (us-east-1, managed Kubernetes, 2 x t3.small) - same manifests also run on minikube

namespace: finsecure (Deployment: 2 replicas, probes, rolling update)

API clients
/health, /validate

Service
LoadBalancer (ELB)

Pod 1 (Ready)
Spring Boot app

Pod 2 (Ready)
Spring Boot app

ConfigMap

Secret

PodDisruptionBudget

logs

metrics

namespace: monitoring

Promtail
reads pod logs

Loki
log store

Grafana
live dashboard

Prometheus
scrapes app + cAdvisor

Alertmanager
alert rules

solid = data flow, dashed = query / scrape; rejected-transaction alert fires when rejection ratio > 50%

Engineer

dashboards, kubectl logs

SMTP inbox
alert + build emails

webhook

AI triage (proposed)
LLM + anomaly detection

suggests cause + action, human approves rollback